PROJECT REPORT

Employee Management System

Human Resource Management System — HRMS

Prepared By	**Eng. Ranim Tayseer Alqrenawi**
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Technology Stack	Python 3.8+ · Tkinter · SQLite3
Project Type	Desktop Application — Human Resource Management
Classification	Academic Project Submission

■ Python 3.8+ ■ Tkinter GUI ■ SQLite3 ■ ReportLab ■ Matplotlib

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1. Introduction

In the modern business environment, the effective management of human resources is a cornerstone of organizational success. Traditional paper-based and semi-digital HR systems often introduce inefficiencies, errors, and communication gaps that hinder productivity. The **Employee Management System (HRMS)** presented in this report is a comprehensive, fully offline desktop application designed to centralize and automate the core functions of human resource management.

Developed using **Python 3.8+** with a Tkinter graphical user interface and SQLite3 as the backend database, this system provides organizations — particularly small to medium-sized enterprises — with a robust, cost-free alternative to expensive cloud-based HR platforms. The system covers the full HR lifecycle: from employee onboarding and attendance tracking, to leave management, payroll processing, and multi-format reporting.

This report documents the project in its entirety, covering objectives, design rationale, technologies, system architecture, use cases, data model, inputs, outputs, and recommendations for future development.

2. Project Objectives

2.1 Primary Objectives

- Provide a centralized platform for managing all employee-related data including personal details, departmental assignments, and employment history.
- Automate attendance tracking through digital check-in and check-out functionality, eliminating manual logbooks.
- Streamline the leave request and approval workflow within a single integrated platform.
- Automate payroll calculations based on attendance records, leave deductions, and configured salary structures.
- Generate comprehensive reports in multiple formats: PDF, Excel (.xlsx), and visual charts (PNG).

2.2 Secondary Objectives

- Implement role-based access control ensuring data security across the organization.
- Support bilingual operation in both Arabic and English for regional business contexts.
- Provide dark mode and theme customization (Light / Dark / Blue) for improved user experience.
- Enable automated email notifications for key HR events via SMTP.
- Include database backup and restore capabilities to ensure business continuity.

3. Rationale for Choosing This Project

The decision to develop this system stems from a clear gap between the needs of small and medium-sized organizations and the HR tools currently available to them. Large-scale commercial platforms such as SAP HCM, Oracle HCM, or Workday are prohibitively expensive and require constant internet connectivity, specialized IT infrastructure, and ongoing licensing fees.

- **Market Gap:** Many organizations still manage HR using spreadsheets, paper forms, or disconnected tools, leading to data inconsistencies and inefficiencies.
- **Accessibility:** A 100% offline solution removes dependency on internet infrastructure — ideal for regions with unreliable connectivity.
- **Cost Elimination:** A fully free desktop application removes financial barriers preventing smaller organizations from adopting proper HR technology.

- **Localization:** Arabic language support and Arabic-language PDF generation address a significant regional requirement that most commercial systems handle poorly.
- **Academic Value:** The project demonstrates system design, relational database modeling, GUI development, and software engineering best practices.

4. Problems in Competing Systems

An analysis of existing HR management tools reveals recurring shortcomings that this system directly addresses:

Problem Area	Description
High Cost	Enterprise platforms (SAP, Oracle, ADP) require expensive licenses, implementation fees, and annual subscriptions inaccessible to SMEs.
Internet Dependency	Most modern SaaS HR platforms require stable internet, making them unusable in offline environments.
Poor Localization	Arabic language support is limited or absent; Arabic-language PDF generation is particularly rare.
Complex Setup	Many systems require dedicated servers, IT personnel, and complex configuration.
Feature Overload	Large-scale platforms overwhelm small teams, reducing usability and increasing training costs.
No Dark Mode	Most HR desktop tools lack modern UI themes, impacting user comfort during extended usage.
Weak Reporting	Competing tools often lack integrated PDF, Excel, and chart reporting simultaneously.
Limited Role Control	Simpler HR tools lack fine-grained permissions, risking unauthorized data access.

5. Unique Features

Feature	Benefit
100% Free & Open	No licensing fees or subscriptions — fully accessible to any organization.
Works Completely Offline	All data stored locally in SQLite; no internet required.
Arabic & English Support	Full bilingual interface with language switching and Arabic PDF generation.
Dark / Light / Blue Themes	Three selectable UI themes for user comfort and accessibility.
Integrated Payroll Automation	Calculates payroll automatically from attendance records.
Recent Activities Feed	Real-time audit feed enhancing transparency and accountability.
Multi-Format Reporting	Exports reports as PDF, Excel (.xlsx), and PNG charts.
Email Notifications (SMTP)	Automated alerts for leave approvals and payroll events.
Database Backup & Restore	Built-in tool for creating and restoring timestamped backups.
Role-Based Access Control	Four distinct user roles with tailored permissions.

6. Technologies Used

Technology	Role in Project
Python 3.8+	Core programming language providing application logic, business rules, and integration.
Tkinter	Built-in Python GUI toolkit for all graphical windows, forms, tables, and dashboards.
SQLite3	Lightweight file-based relational database; all persistent storage; no server required.
ReportLab	Python library for programmatic PDF generation including Arabic-language reports.
Matplotlib	Charting library for statistical visualizations exported as PNG.
Pandas	Data manipulation for processing payroll and attendance datasets for Excel export.
OpenPyXL	Library for creating and writing .xlsx Excel files from Python.
smtplib / SMTP	Standard library module for sending automated email notifications.

7. Project Structure

Module / Directory	Contents & Purpose
database/	db_manager.py: all DB connections and CRUD operations. schema.sql: table structures and seed data.
models/	One module per entity: employee, department, attendance, leave, payroll, activity. Each encapsulates domain business logic.
views/	Ten Tkinter GUI windows: dashboard, employee form, department manager, attendance, leave management, payroll, reports, settings, user management, backup tool.
utils/	session.py, config.py, email_utils.py, validators.py — supporting utilities for session state, configuration, email, and input validation.
main.py	Application entry point: initializes the database, loads configuration, and launches the main window.

8. Use Case Diagram — Textual Description

8.1 Actors

Actor	Description
Admin	Full system access. Manages users, roles, system configuration, backup and restore.
HR Manager	Manages employees, processes leave requests, runs payroll, generates reports.
Department Manager	Views and manages only their own team. Approves or rejects team leave requests.
Employee	Views personal profile, logs attendance (check-in/out), submits leave requests.

8.2 Use Cases by Actor

Admin

- Manage Users (Create, Edit, Deactivate)
- Assign Roles
- Backup & Restore Database
- Configure System Settings (language, theme, SMTP)
- All HR Manager features

HR Manager

- Add / Edit / Delete Employees
- Search and Filter Records
- Manage Departments
- Review and Approve / Reject Leave Requests
- Process Monthly Payroll
- Generate PDF & Excel Reports
- Send Email Notifications

Department Manager

- View Employees in Own Department
- Approve / Reject Team Leave Requests
- View Team Attendance Records

Employee

- View Personal Profile
- Check In / Check Out
- Submit Leave Request
- Track Own Leave Status

9. Entity-Relationship Diagram

The database consists of **eight tables** with clearly defined relationships. The diagram below was generated from the actual database schema and illustrates all entities, attributes, and their interconnections.

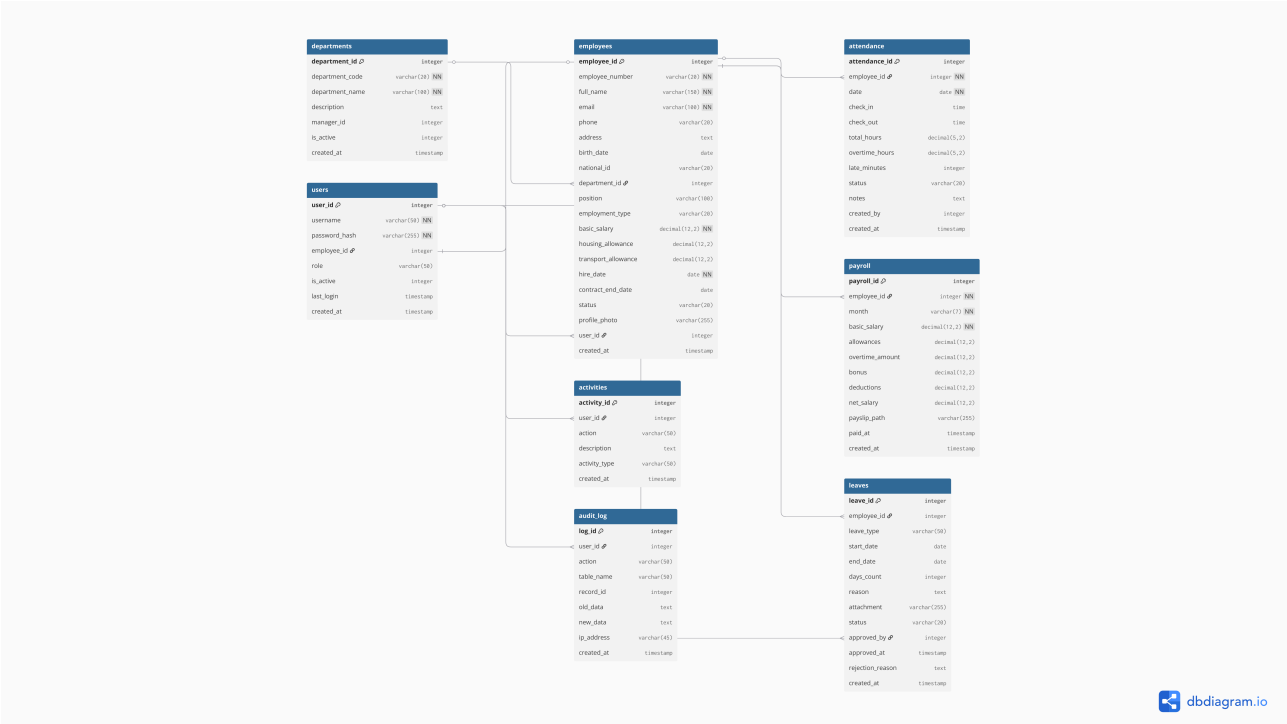

Figure 1: HRMS Entity-Relationship Diagram

9.1 Entities and Their Attributes

Table	Key Fields & Count
departments	department_id (PK), department_code, department_name, description, manager_id (FK), is_active, created_at — 7 fields
employees	employee_id (PK), employee_number, full_name, email, phone, address, birth_date, national_id, department_id (FK), position, employment_type, basic_salary, housing_allowance, transport_allowance, hire_date, contract_end_date, status, profile_photo, user_id (FK), created_at — 20 fields
users	user_id (PK), username, password_hash, employee_id (FK), role, is_active, last_login, created_at — 8 fields
attendance	attendance_id (PK), employee_id (FK), date, check_in, check_out, total_hours, overtime_hours, late_minutes, status, notes, created_by (FK), created_at — 12 fields
leaves	leave_id (PK), employee_id (FK), leave_type, start_date, end_date, days_count, reason, attachment, status, approved_by (FK), approved_at, rejection_reason, created_at — 13 fields
payroll	payroll_id (PK), employee_id (FK), month, basic_salary, allowances, overtime_amount, bonus, deductions, net_salary, payslip_path, paid_at, created_at — 12 fields
activities	activity_id (PK), user_id (FK), action, description, activity_type, created_at — 6 fields

audit_log	log_id (PK), user_id (FK), action, table_name, record_id, old_data, new_data, ip_address, created_at — 9 fields

9.2 Relationships

Relationship	Type	Description
departments → employees	1 : Many	Each department contains many employees; each employee belongs to one department.
employees → users	1 : 1	Each employee has exactly one system user account for authentication.
employees → attendance	1 : Many	Each employee accumulates multiple daily attendance records.
employees → leaves	1 : Many	Each employee can submit multiple leave requests over time.
employees → payroll	1 : Many	Each employee has monthly payroll records generated over time.
users → activities	1 : Many	All user actions generate activity log entries.
users → audit_log	1 : Many	All data changes by a user are tracked in the audit log.

10. System Inputs and Outputs

10.1 System Inputs

Input Category	Fields / Data Collected
Login	Username, Password
Employee Registration	Full Name, Email, Phone, Department, Position, Base Salary, Housing & Transport Allowances, Hire Date, National ID, Address, Birth Date, Gender, Emergency Contact, Bank Account
Attendance	Employee selection, Check-In timestamp, Check-Out timestamp
Leave Request	Leave Type (Annual / Sick / Unpaid / Other), Start Date, End Date, Reason, Attachment
Leave Decision	Approver, Approve or Reject action, Optional rejection reason
Payroll Processing	Target Month and Year selection
Settings	Display Language, UI Theme, SMTP Server, Email Credentials
User Management	Username, Role, Account activation status
Backup & Restore	Target directory path for backup; source file for restore

10.2 System Outputs

Output Type	Description
Dashboard Statistics	Live summary: total employees, present today, pending leaves, monthly payroll total.
Employee Table	Searchable, filterable employee grid with status indicators.
Attendance Records	Daily log per employee with hours worked, late minutes, and status.
Leave Requests Table	All requests with status badges (Pending / Approved / Rejected).
Payroll Table	Breakdown: base salary, allowances, overtime, deductions, net pay.
PDF Reports	Professionally formatted reports for Employees, Attendance, Payroll — Arabic supported.
Excel Files (.xlsx)	Payroll data exported to spreadsheet for external analysis.
PNG Charts	Bar/pie charts of employee distribution across departments.
Email Notifications	Automated emails for leave status changes and payroll events.
Activity Feed	Real-time sidebar log of recent system actions.
Audit Log	Detailed change history with old/new values for sensitive modifications.
Database Backup File	Timestamped .db file for disaster recovery and migration.

11. Conclusion and Recommendations

11.1 Conclusion

The **Employee Management System (HRMS)** successfully delivers a feature-complete, offline-capable, and cost-free solution to the real-world challenges of human resource management in small and medium-sized organizations. By integrating employee records, attendance, leave workflows, payroll automation, and multi-format reporting into a single cohesive desktop application, the system eliminates the fragmentation and manual errors that characterize paper-based HR operations.

The adoption of Python, Tkinter, and SQLite ensures the application is lightweight, portable, and deployable without server infrastructure. Arabic language support, PDF generation, dark mode theming, role-based access control, and email notifications elevate the system well beyond a basic prototype — it represents a genuinely deployable production solution.

The modular architecture — with clearly separated database, model, view, and utility layers — ensures the codebase is maintainable and extensible. The audit logging and activity feed add governance and accountability that many competing low-cost tools entirely lack.

11.2 Recommendations for Future Development

1. **Web-Based Interface:** Develop a companion web app using Flask or Django for remote multi-device access.
2. **Mobile Application:** Build an Android/iOS app enabling employees to check in/out and submit leaves from mobile devices.
3. **Advanced Analytics:** Add turnover rates, absenteeism trends, and predictive modeling using Pandas and Matplotlib.
4. **Biometric Integration:** Connect to fingerprint or facial recognition hardware for tamper-proof attendance.
5. **Cloud Backup:** Add optional encrypted backup via AWS S3 or Google Drive.
6. **REST API Layer:** Expose an API for integration with payroll processors, accounting software, or ERP systems.
7. **Database Scaling:** Migrate from SQLite to PostgreSQL for larger datasets and concurrent multi-user access.
8. **Expanded Languages:** Add French, Urdu, or Turkish to serve additional regional markets.

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Eng. Ranim Tayseer Alqrenawi.

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